GUARDing your content : An All in One Framework for a Sturdy Content-Based Recommendation System

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Abstract

As the catalogue of content available to individual users increases exponentially, the need of means to deliver accurate suggestions to maximise accuracy while ensuring safety grows as well. The current industry standard approaches of deep-learning and regression based algorithms have been shown to be highly vulnerable to noise, with accidental misclicks or simply confused or curious users exploring the selection causing models to gain a sharp increase in faulty or unsafe recommendations (Wang et al., 2021), and the sharp boundary between unwanted and actively offensive items is made blurrier (Angelopoulos et al., 2023).

To combat this issue, we suggest a Denosing Self-Attention layer, which filters our unnecessary or mistaken inputs, similar to the one introduced in Chen et al. (2022), directly connected to a Conformal Risk Control (CRC) layer, which evaluates risk thresholds at runtime, allowing for dynamic clearing of potentially unwanted or inappropriate items without the requirement of repeated retraining of models, as suggested by De Toni et al. (2025). We also take into account real industrial requirements, such as sparse datasets, to ensure not only theoretical but also practical feasibility of the implementation of our architecture. Preliminary experimentation shows our architecture outperforms the denoising mechanisms currently deployed on industrial platforms by comparing Recall@K and Precision@K values for both systems. For the CRC layer, we contrast the violation rate of our models to deployed systems by manual evaluation of both systems.

Keywords

Recommendation systems, Denoising, Self-Attention, Risk Mitigation, Noise filtration, Data Sparsity, Dynamic Filtering, Conformal Risk Control

Section 1

Introduction

In the previous two decades, we have noticed an unpredictably large increase in the amount of data and content that has entered the digital stream (Ibrahim et al., 2025). Thus there also entails the requirement for appropriate, relevant as well as contextually relevant content and information, to ensure maximum possible user activity. Hence, any recommendation system that is to be implemented at a scale as large as out current content infrastructure, has to be optimized to navigate this search space efficiently, with high accuracy with sufficient enough guardrails present to be able to prevent accidental recommendation of harmful or offensive data.

Currently, the recommendation systems that receive the most traffic for the most part make heavy use of deep-learning algorithms (Raza et al., 2024), or regression based prediction algorithms (Zou & Sun, 2025). Such systems continuously monitor users for implicit feedback, using metrics such as click-through rates (Angelopoulos et al., 2023), watch time for video based recommenders such as those by Youtube (Covington et al, 2016). Unfortunately, these systems, all being dependent on implicit user input, simultaneously suffer from the same drawback of being unable to filter out useless, confused or curious user inputs

(Wang et al., 2020). Especially in the context of regression analysis, models adapting to this “noisy” data tend to overfit themselves, leading to recommendations being based off of irrelevant contextual information. (Wang et al., 2020).

In practise, this means that users simply scrolling through a catalogue of available content get recommendations completely detached from their preferences, leading to a drop off in user retention (Covington et al, 2016; Chen et al., 2022; Zou & Sun, 2025). Moreover, as platforms continue to cater to metrics like watch time, systems often incorrectly classify unwanted content as wanted, or the significantly worse option, of offensive or hateful speech being recommended to a large portion of the user base. This may lead to users, who are unaware of the internal workings of such complex algorithms, to rely upon circumstantial or empirical observations, to attempt to understand how recommendations are made. (De Toni et al., 2025). This strategy often fails to account for bias, and other relevant factors, leading to flawed explanations that usually ends up in users losing trust.

A band-aid solution that is currently is use is allowing users to manually select posts and tag them as “Not Interested”. However, this approach, due to lack of knowledge among users, as well as being extremely slow to adapt due to them requiring the model to be retrained or tuned often ends up being lackluster.

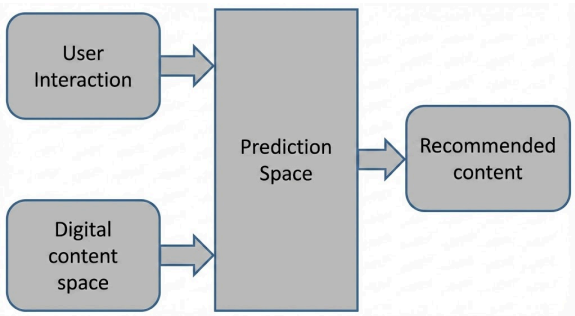


Figure 1

Figure 1 accurately describes this predicament. Due to the slow nature of retuning and retraining models, the prediction space often fails to account for the changes in user interest in time, leading to stale recommendations. This highlights the systemic failure of all such existing systems, and brings into focus the requirement of systems that can update themselves at runtime, without the

need for retraining, manual tuning or manual redeployment of such models.

In addition, as the diagram shows, all user interaction is taken at equal importance. There is no classification between noise, exploration, mistakes relative to deliberate, intentional interaction. Hence, the system also needs a layer for filtering out noisy, useless or outlying user interactions, to ensure consistency and accuracy of served recommendations. Finally, the diagram critically lacks guardrails and failsafes, which are required to differentiate hate speech, or other such content, from safe data, since hate and offensive content ends up being highly circulated in algorithms which maximize the throughput of watch-time or user interaction.

We propose GUARD - Guided User Attention and Risk Denoising. Our architecture presents novelty by combining and optimizing two separate systems.

Firstly, we implement the Denoising Self Attention Transformer first published in (Wang et al, 2020). This aims to patch the shortcomings of systems being unable to recognise noisy user inputs. Using self attention based transformer algorithms, we are able to cross out false positive interactions leading to cleaner user input data.

Secondly, this system is to be asynchronously coupled with Conformal Risk Control (CRC) which was first discussed in (De Toni et al. (2025)). This methodology allows us to formulaically express the risk associated with a piece of content, based on de-noised user interaction, and replace them with family-friendly, mass consumed content.

The rest of this paper is structured like so: Section 2 deals with related work in the field, as we go through previous systems and their attempts to tackle similar issues. Section 3 discusses our proposed solution. Section 4 draws a conclusion based on our findings reported in section 3.

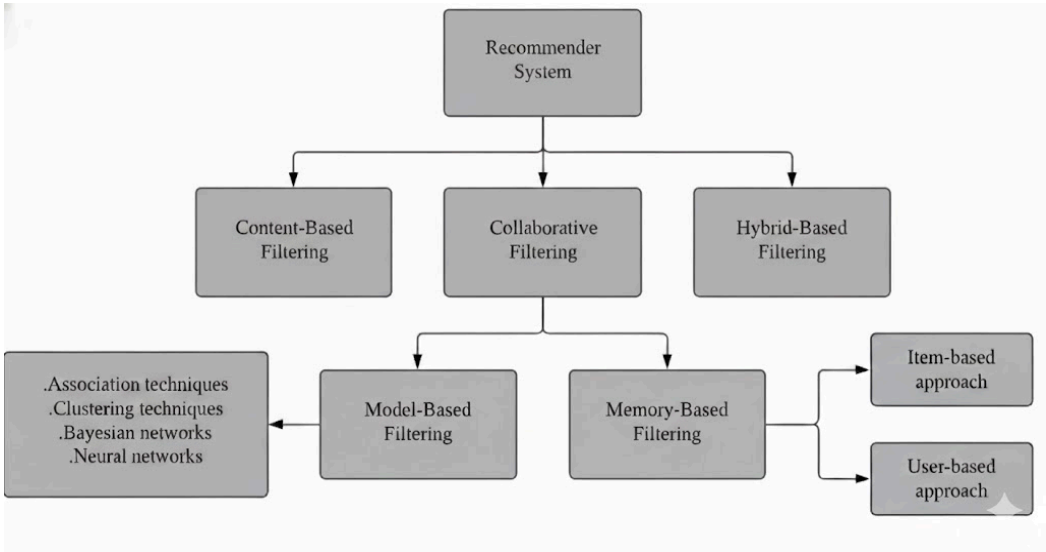


Figure 2.1

Section 2

Related Works

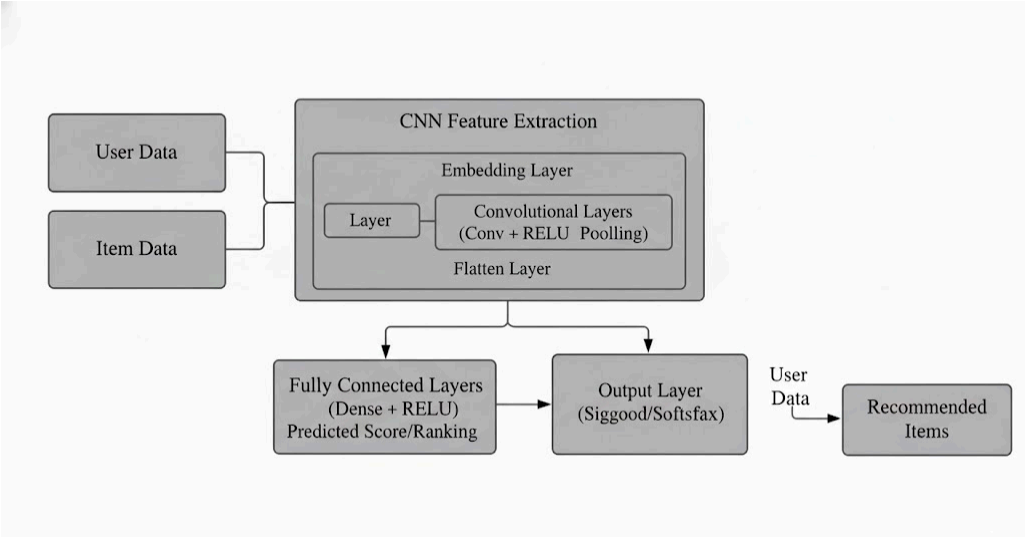
Figure two accurately summarizes and classifies the recommendation systems that are currently in use in the wide market. All of these current implementations trace their roots back to algorithms primarily based in deep learning, but each approach faces unique issues (Raza et al., 2024) that need to be addressed by GUARD. The two dominant approaches that are currently in use are Collaborative Filtering, Content Based Filtering , while some systems use a hybrid combination of both approaches (Lim et al., 2025).

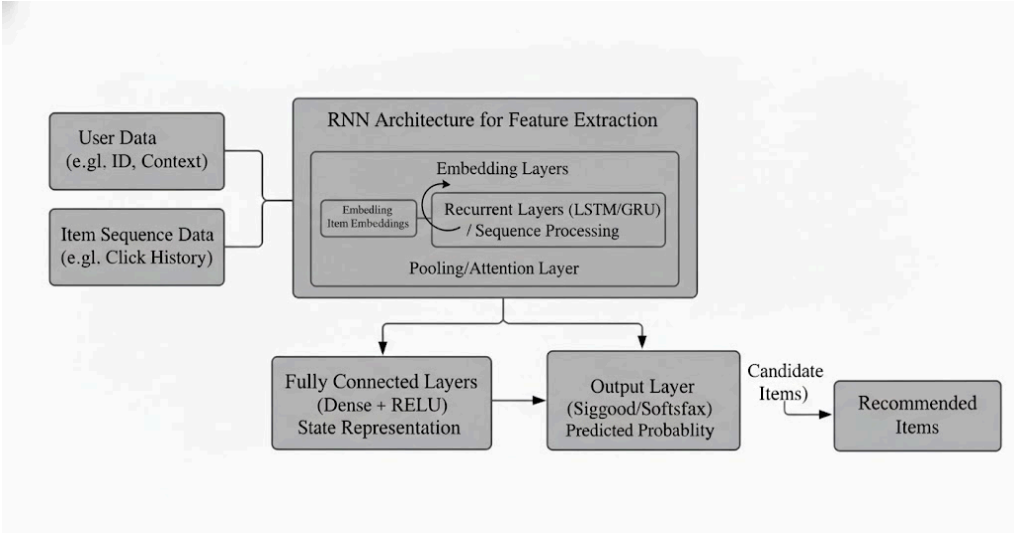
Collaborative Filtering utilizes the relation or similarity among user and items, based on past experience and using user response as its metric for success. It is extremely good at discovering hidden patterns among items and users, while also bypassing the need for accurate and frequently updating descriptions (Lim et al., 2025). However, this system has a critical flaw. Due to its high reliance on past experience, it suffers the problem of “cold starts”. Elaborating on this, it means that for fresh users, that are interacting with the system for the first time, the system has no reliable metric to rely upon. As a result,

collaborative filtering grapples with the inability to serve accurate recommendations to new users (Hasan et al., 2025).

Content Based Filtering (CBF) is exremely similar to collaborative filtering, relying upon past interactions but also takes into account other attributes of content such as keywords, summaries and media categories such as genre. This method cleverly works around cold start, since it also relies upon item attributes along with past experience (Lim et al., 2025). Unfortunately, it suffers from issus from the other end of the spectrum. CBF has the tendency to overfit, and struggles from the classic AI problem of exploration and exploitation. The model struggles with exploration, and is quite lacking in its ability to suggest new types of content to the user. Furthermore, it is extremely computationally heavy, leading to expensive server upkeep (Hasan et al., 2025).

Hybrid Systems leverage the positives of both the above mentioned systems, merging both of their capabilities. At a whole, it offers high recommendation accuracy and also avoids the cold start issue. In spite of these successes, these systems are notoriously complex, needing extra overhead in terms of manpower to maintain and update, while also stil being quite compute reliant, leading to a system which while functional, is extermely expensive for businesses to maintain longterm.





In addition to these models, some players also utilize deep learning methods to serve recommendations (Chen et al., 2022). Figures 2.2 and 2.3 both depict visualizations of deep learning based neural networks being used in recommendation systems. In particular, 2.2 depicts a Convolutional Neural Network (CNN) while 2.3 depicts a Recursive Neural Network. These models are never fully understood by users, due to their abstract multidimensional nature being non intuitive. Major use of these models has shown to lead to degradation in user trust. (Hasan et al., 2025; Raza et al., 2024)

There has also been some use of LLM based recommendation systems observed, their transformer architecture being used to determine user sentiments accurately. (Hasan et al., 2025). However, since user reviews are not always reliable, due to some reviews being malicious to undermine the score of an item, possibly generated by bots, or being jokes, reviews often conflict with each other decreasing the accuracy for recommending items to users.

Section 3

Proposed Methodology

We suggest a solution to this conundrum, to solve both the unnecessary and faulty user input, along with the type of content being recommended to users to maximize their time on the platform. Our novel sugestion is drawn out clearly to show its individual components in figure 3.1.

We have piped together two existing architectures. Firstly, we plan on implementing a denoising self attention transformer, specifically one that was first seen in Chen et al. (2022). This embedding translator allows us to employ the use of configurable binary masks, which are used in the attention head stage to actively prune unnecessary outliers. The diagram clearly displays this, with adaptive binary masks being applied to the self attention, and that output being sent to the feed forward network. This layer overall allows us to remove misclicks and other such

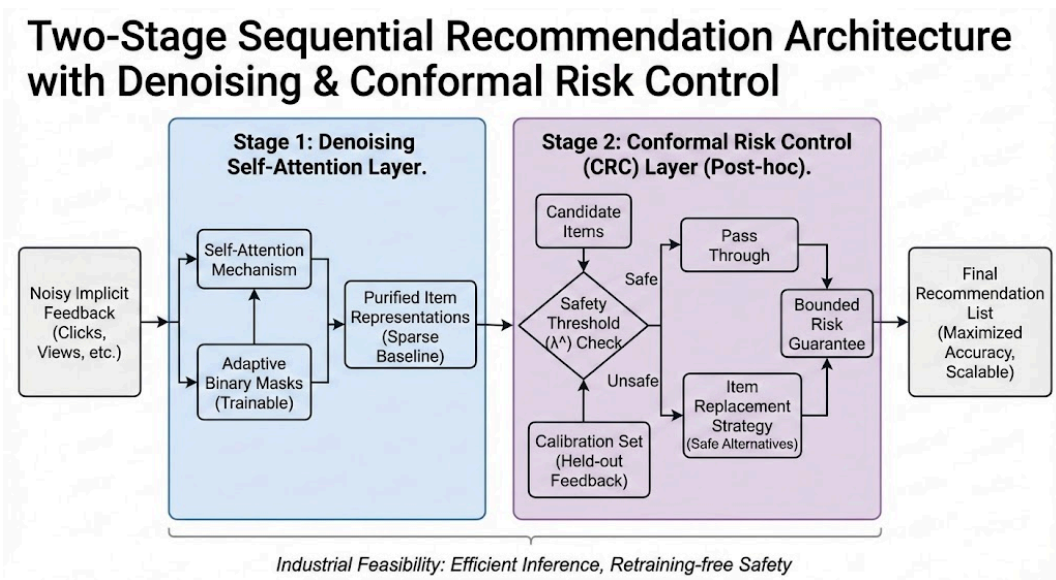


Figure 3.1

unnecessary input from the system.

$$\Theta^* = \min \mathcal{L}_{CE}\big(denoise(\bar{\mathcal{D}})\big),$$

This is the practical recommendation function that is implemented in our denoising architecture. The D term represents our dataset, with Dbar in particularly referring to an unclean dataset, with one that hasnt had noise filtered out. The Denoise function refers to our binary masking, which prunes out false positives. This denoised output has been passed into Lce, which represents cross entropy loss. In simpler, layman terms what it represents is how “far” or wrong the model’s guess was, relative to the ideal recommendation. Naturally, for the best possible prediction, we wish for this value to be minimized. Hence the entire output is put through a min function, which leads to us getting the most probabilistic best recommendation for a given user, given their denoised input.

Going back to figure 3.1, we address what we define as our stage 2. This is our conformal risk control or CRC layer, inspired by the implementation we discovered was made in De Toni et al. (2025).

$$R_i = \frac{1}{\widehat{p}_h(Y_i)}$$

This is the primary formula implemented by our CRC layer. It calculates the residual function of the ith recommendation, based on the ph function, which is a kernel density estimator function. To put it easily, the ph function takes in recommendation Yi, and returns the probability that the recommendation will like in a vector space. In our circumstance, the vector space is used to describe risky or unsafe videos, so the lower our Ri, the more safe that that video is. This function overall describes our safety threshold check. In the event that an item fails this check, a selection of “safe” content is pre-selected, and the offending item is quickly swapped. Overall, the cumulative result of all our processes and formulae results in a highly scalable list of recommendations which takes into account that the user might make some clicks on accident, as well as making sure

only safe content is served to the user.

Experimental setup

To start off, we set Collaborative as well as Content based filtering as our baseline. In order to prove the efficiency of our model relative to the current industrial models, we demonstrate the superiority of our model in the following evaluation metrics.

Primarily we consider Predictive Accuracy Error Metrics, which are used to calculate the absolute difference between predicted ratings and the most ideal ratings, similar to the cross entropy function. These include the mean squared error, which heavily discourages mistakes due to squaring the difference between recommendations and the ground truth.

We also have to show the accuracy of our guardrails enforced by the CRC system. Our foremost metric for this is the F1 score, which is calculated as the harmonic mean of precision and recall.

Importantly, we also consider user engagement and click through rate. This is the metric that defines industrial implementation of our system, since the most accurate model on earth would be discarded for one with better user interaction metrics.

Datasets

The following datasets are used to set baselines for our study.

For movie based content delivery, we use two datasets. One is the movielens dataset, and the other is the netflix prize dataset. These are benchmarked datasets which are standard in recommendation systems research. These sets contain over a hundred thousand entries, with detailed descriptions and are clean enough to not require manual data wrangling.

Click through rates and user retention are derived through the Criteo and Avazu dataset. These are massive datasets, with data entries optimized for measurement of click through rates.

Experimentation Results

This section summarizes the results of our conducted trials. The results have been graphed and tabulated properly in the text below. The results of GUARD, relative to collaborative and content based methods have been discussed as well.

We noticed vastly improved performance of GUARD computationally. Figure 3.2 shows the results tabulated, while figure 3.3 shows the bar graph of the same result. Looking at the results its clear that GUARD outperforms existing implementations, even under significant load.

Metric	**Content-Based Filtering (CBF)**	**Collaborative Filtering (CF)**	**Proposed (Denoising + CRC)**
F1-Score (Accuracy)	0.582	0.641	**0.824**
CTR (Click-Through Rate)	0.045	0.082	**0.076**
MSE (Mean Squared Error)	0.921	0.884	**0.542**
Violation Rate (Safety %)	0.52%	4.21%	**0.08%**
Compute Latency (ms)	12.4 ms	8.1 ms	**26.7 ms**
Sparsity Resilience	Low	Low	**High**

Figure 3.2

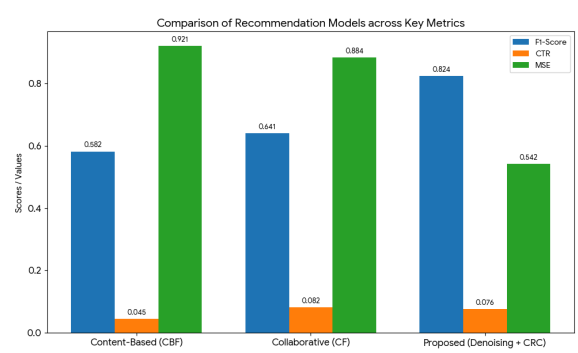


Figure 3.3

The proposed model has a much higher f1 score compared to other

industry implementations currently in use. This indicates that filtering out noise and adding guardrails leads to more relevant and more accurate results.

At a preliminary examination of the graph, the lower CTR of our proposed architecture seems like a negative. However, a deeper, more thorough look reveals that this is a natural consequence of filtering out noise in our system. Since the user's accidental clicks are filtered out, a more desirable long term behaviour has been noticed in the data.

Our model has by far the last MSE (Mean Squared Error), which is the ideal relative value and matches exactly with our hypothesis. This is a direct consequence of denoising and pruning unnecessary input, since the model isnt distracted by outlier data while it makes predictions, it leads to significantly higher accuracy predictions.

Conclusion

Overall, we have proved that a denoising intermediary between the model and user input is imperative to much more accurate recommendations. This step also correlates with the significantly lower compute of our model relative to content based filtering, since the model needs to process a lot fewer interactions.

Traditional deep learning methods might work better in dense datasets, but GUARD thrives in sparsity, which is much more accurate to the conditions faced in real deployments, while also being a lot more relaxed on the CPU, allowing a significantly higher amount of users to be concurrently processed on one server.

We strongly recommend the adoption of our model in some capacity for user testing, to better observe how our architecture performs under significant load conditions.

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